

Product Laws of ExponentsQuestion Bank - Additional 3.01 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 and 2, write the product as a repeated multiplication expression, then as a single power.**

1. $4^2 \cdot 4^5$

$$(4 \cdot 4)(4 \cdot 4 \cdot 4 \cdot 4 \cdot 4) = 4^7$$

2. $7^2 \cdot 7^1 \cdot 7^3$

$$7^6$$

$$2 + 1 + 3 = 6.$$

3. Evaluate each power, then multiply. $5^3 \cdot 5^2$

$$125 \cdot 25 = 3,125$$

Also 5^5 .

4. Apply the product of powers law, then evaluate the power. $(-6)^2 \cdot (-6)^7$

$$(-6)^9 = -10,077,696$$

Nine negative factors: negative.

5. Leeland says he can evaluate $7^3 \cdot 3^7$ using the product of powers law. Is he correct? Explain.

No — the bases are different (7 and 3), so the law does not apply. The value is $343 \cdot 2,187 = 750,141$, found by evaluating each power.

For questions 6 through 8, use the power of a product law.

6. Write $(7 \cdot 8)^2$ as a repeated multiplication expression, group the repeated factors, then write it in the form $a^m \cdot b^n$.

$$(7 \cdot 8)(7 \cdot 8) = (7 \cdot 7)(8 \cdot 8) = 7^2 \cdot 8^2$$

Value 3,136 either way.

7. Evaluate inside the parentheses first, then apply the exponent. $(-3 \cdot 6)^4$

$$(-18)^4 = 104,976$$

Math Nation's key prints 104,796 for this item; it is 104,976.

8. Apply the power of a product law, then evaluate. $(2 \cdot (-9))^3$

$$2^3 \cdot (-9)^3 = 8 \cdot (-729) = -5,832$$

For questions 9 and 10, use the power of a power law or the identity law.

9. Write $[(5)^4]^4$ as a single power, then give its value.

$$5^{16} = 152,587,890,625$$

10. Rewrite $[(2)^5]^2$ using a law of exponents, name the law, then evaluate.

$$2^{10} = 1,024 \quad \text{— power of a power}$$

Math Nation's key stops at 2^{10} ; the question asks for the value.

Working backwards.

11. Find the value of n that makes each statement true.

a. $3^n \cdot 3^4 = 3^9$

$$n = 5$$

b. $(2^n)^3 = 2^{12}$

$$n = 4$$

c. $\left(\frac{1}{2}\right)^n = \frac{1}{32}$

$$n = 5$$

12. Which expression is equivalent to $(3^4)^2$?

☒ A. 3^8

☐ B. 3^6

☐ C. 9^8

☐ D. 3^{16}

A

Multiply the exponents.

13. Select ALL of the expressions that are equivalent to 10^6 .

☒ **A.** $10^3 \cdot 10^3$

☒ **B.** $(10^2)^3$

☒ **C.** $10^2 \cdot 10^4$

☒ **D.** 100^3

☐ **E.** $10^2 \cdot 10^3$

☐ **F.** $(10^3)^3$

A, B, C and D

D: $100^3 = 1,000,000 = 10^6$.